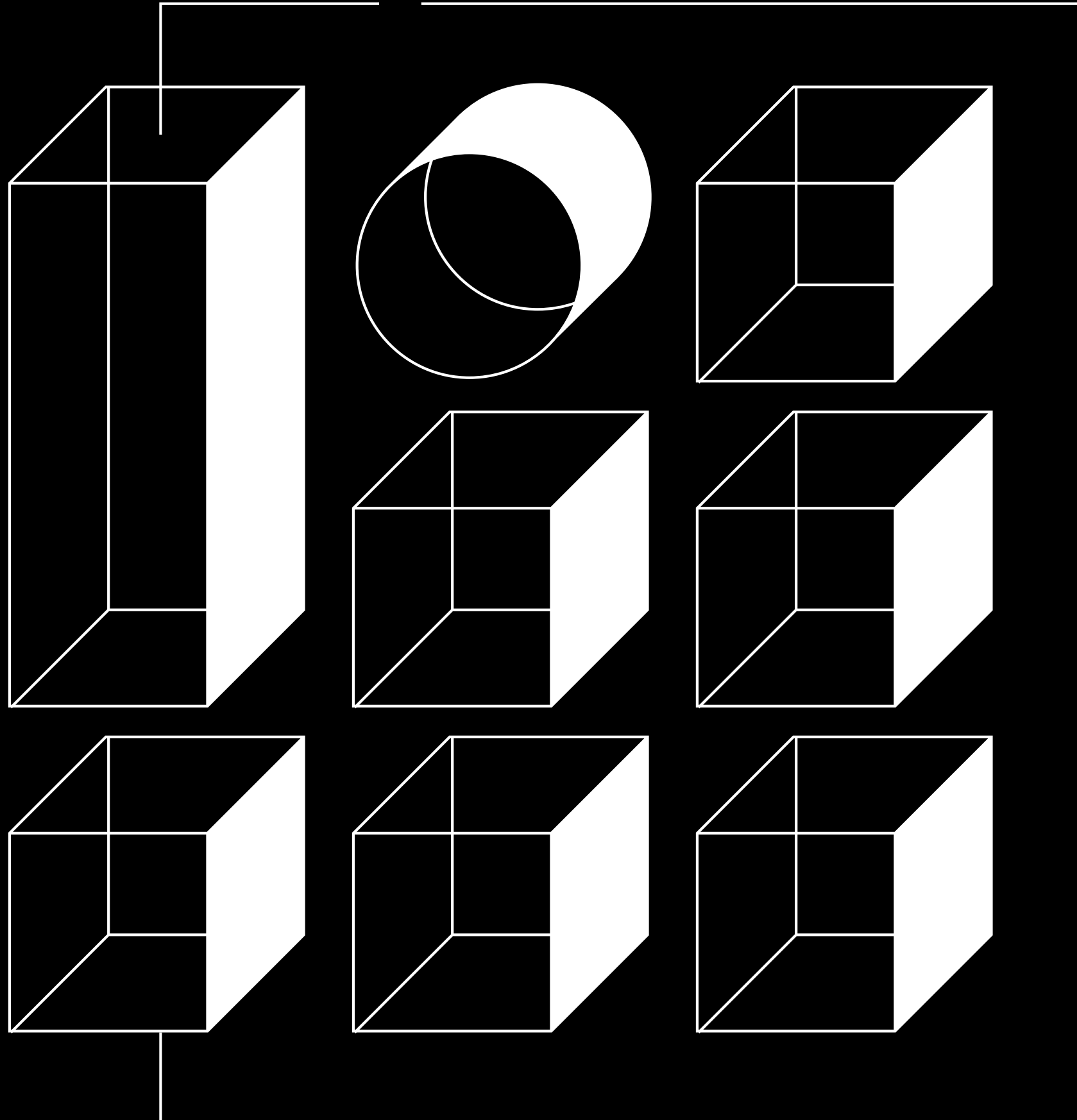




Choosing the Right AI
AI Approach &
Designing for
Reality



Why This Session Matters

Now comes the hardest part: **Deciding what to build** and how to deploy it in the real world.



The Core Question

Key considerations for AI discussions

- Is this even an AI problem?
- If yes, what kind of AI?
- What are the constraints?
- Wrong answers waste time and money.

First Decision

Rules vs Learning in ML

Rules Work Well

Rules are effective when the **logic is clear**, they remain stable over time, and edge cases are limited, allowing for straightforward implementation and maintenance without excessive complexity.

Consider Machine Learning

If the complexity of rules **increases significantly**, it is advisable to explore machine learning approaches that can handle intricate patterns and adapt to evolving data scenarios more efficiently.

Rules vs Learning

Understanding when to apply rules effectively

Logic is clear

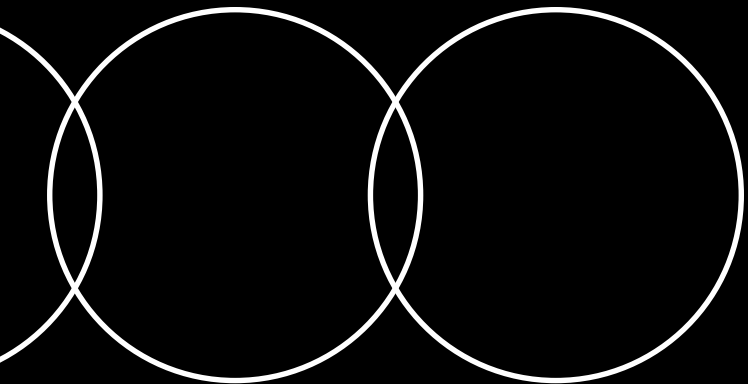
Rules are most effective when the underlying logic is **easily** understood and can be clearly defined without ambiguity.

Rules are stable

Stability in rules ensures consistency; they should remain applicable over time without frequent changes or **modifications**.

Edge cases are limited

When the number of edge cases is small, rules can manage exceptions efficiently without becoming overly complex or **cumbersome**.



Rules vs Learning

Exploring key examples of rule-based solutions

Password Validation

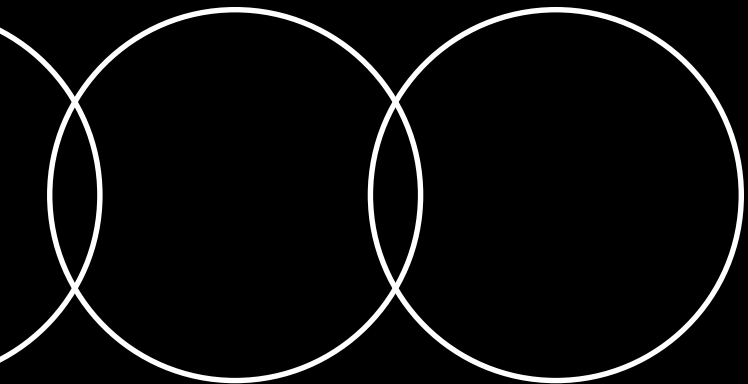
Password validation is a straightforward task that can be handled effectively using clear and stable rules for security.

Tax Calculation

Tax calculation follows established guidelines, making it suitable for rule-based approaches, ensuring accuracy and consistency in outcomes.

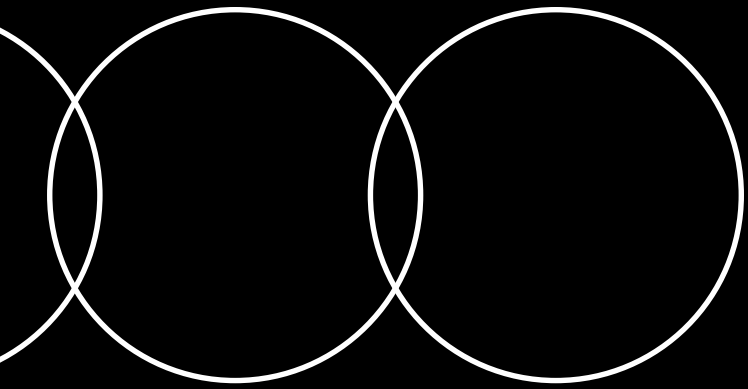
Complexity Consideration

If the complexity of rules becomes unmanageable, it's advisable to explore machine learning solutions for more flexible problem-solving.



Classical Machine Learning

Key conditions for effective model application



Structured Data

Classical ML performs best when data is organized in structured formats like tables, enhancing model training and evaluation efficiency.

Clear Target

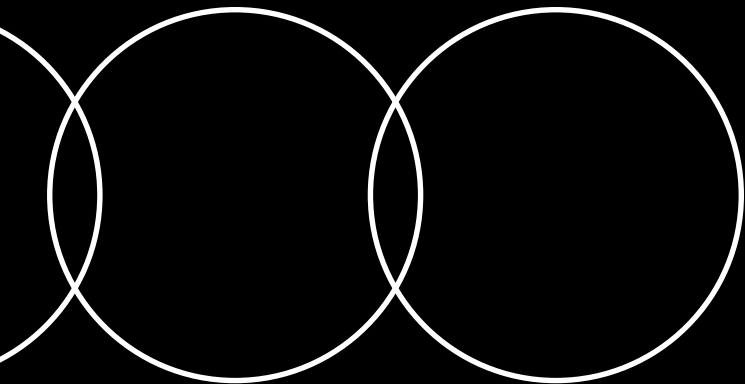
A clear target variable is essential for successful classical ML, guiding the model to learn from inputs and make predictions.

Defined Problem

This approach is well-suited for classification or regression problems, allowing for accurate analysis and actionable insights from data.

Classical ML Examples

Understanding practical applications
of classical machine learning



Credit Risk

Credit risk scoring involves assessing an applicant's likelihood of defaulting on a loan based on historical data and behaviors.

Price Prediction

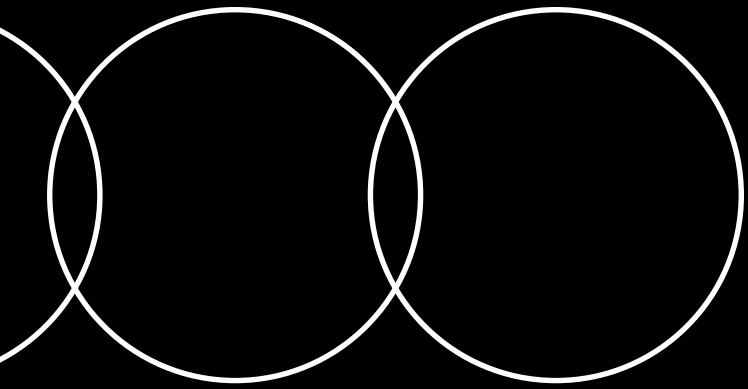
Price prediction uses historical data to forecast future values, allowing businesses to make informed decisions on pricing strategies and inventory.

Churn Prediction

Churn prediction identifies customers likely to leave a service, enabling targeted strategies to retain them and improve overall customer satisfaction.

Classical ML Models

Understanding different types of classical machine learning models



Linear Regression

Linear regression is a statistical method for modeling relationships between variables, predicting outcomes based on linear associations.

Logistic Regression

Logistic regression is used for binary classification problems, estimating probabilities of outcomes using a logistic function approach.

Tree-based Models

Tree-based models use decision trees to make predictions, effective for handling complex datasets with non-linear relationships and interactions.

Traditional NLP Focus

Key Areas of Application

TEXT CLASSIFICATION

This process involves categorizing text into predefined classes based on its content and contextual information.

INFORMATION EXTRACTION

It focuses on identifying and extracting structured information from unstructured text, such as names and dates.

SEARCH AND RANKING

This technique enhances information retrieval by ranking documents based on relevance to the user's query.

Techniques in NLP

Key Methods for Text Analysis

TF-IDF

TF-IDF stands for Term Frequency-Inverse Document Frequency, highlighting words' importance in a document relative to the entire corpus.

BAG OF WORDS

The Bag of Words model simplifies text into word counts, ignoring grammar but retaining the overall meaning for analysis.

WORD EMBEDDINGS

Word embeddings represent words in a continuous vector space, capturing semantic relationships and context for improved language understanding.

Examples in NLP

Applications of traditional methodologies

SENTIMENT ANALYSIS

Sentiment analysis determines the emotional tone behind a series of words, helping to understand user opinions and feelings.

SPAM FILTERING

Spam filtering uses traditional NLP methods to identify and filter unwanted or harmful emails, improving inbox management for users.

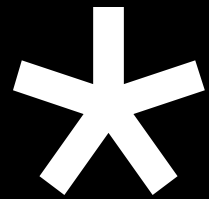
SOURCE

Source: This information is derived from the Stanford NLP book, a comprehensive resource on information retrieval.

Large Language Models

Deep Learning for Natural Language Tasks

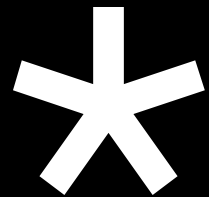
LLMs are advanced models that leverage extensive text data, enabling **natural language understanding**, **text generation**, and **generalization across diverse tasks** to enhance user interactions and automate processes.



Large Language Models

Key capabilities of LLMs explained

LLMs excel at **natural language understanding**, enabling them to accurately process and comprehend text, **text generation**, producing coherent and contextually relevant content, and **generalization across tasks**, adapting knowledge effectively.



When LLMs Are Appropriate

Use LLMs for specific scenarios

Complex Language

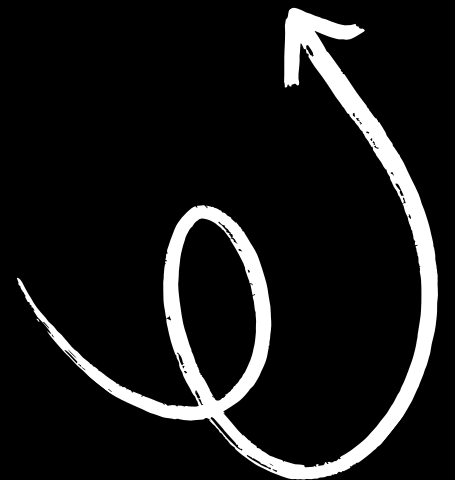
LLMs excel in tasks that require **complex language understanding**, allowing for nuanced interpretations and responses that simpler models cannot provide effectively in conversational contexts.

Undefined Rules

When the rules of a task are **impossible to define**, LLMs offer the flexibility needed to adapt to varying inputs and contexts, ensuring relevant outputs.

Limited Data

In scenarios where **training data is limited**, LLMs can leverage their pre-trained knowledge to generate insightful responses, filling in gaps that traditional models may struggle with.



Examples of LLMs

Applications in various fields

Conversational Assistants

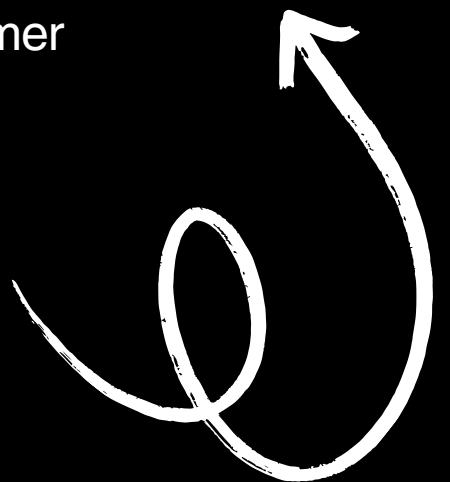
LLMs excel as conversational assistants, providing **contextually relevant responses** in real-time, enhancing user interaction and engagement through natural, flowing dialogue and understanding user intent effectively.

Legal Document Analysis

Leveraging LLMs for legal document analysis enables **efficient extraction of insights** and key information, simplifying complex legal texts and supporting professionals with faster, more accurate evaluations of documentation.

Customer Support Bots

Implementing LLMs in customer support bots enhances **service quality and responsiveness**, allowing businesses to automate inquiries and provide personalized solutions, significantly improving overall customer satisfaction and efficiency.



When LLMs Are the Wrong Choice

01

Simple Classification

Avoid LLMs for straightforward classification tasks.

03

Critical Cost

Latency or cost considerations are paramount here.

02

Structured Data

Dominance of structured data eliminates LLM benefits.

04

Deterministic Behavior

Predictable outcomes are necessary for success.

Interview- Style Example

PROBLEM

"We want to detect **fraudulent transactions**." This problem requires a thoughtful approach to ensure accuracy and effectiveness in identifying suspicious activities without introducing false positives.

CORRECT FRAMING

Utilize **structured data**, focusing on a **binary outcome** through **supervised classification** techniques. This methodology ensures reliable predictions while maintaining a clear distinction between legitimate and fraudulent transactions.

Architecture Patterns

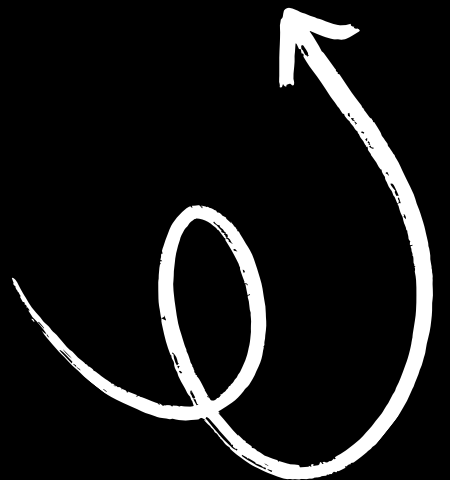
Exploring common architecture patterns for NLP and LLMs can enhance understanding of efficient implementations and inform decisions for various applications in artificial intelligence.



Why Cost & Latency Matter

Balancing performance and affordability

A model can be **accurate** and **elegant** yet still be **too slow** or **too expensive**. Achieving production success requires a careful balance between these critical factors.



Cost Drivers in AI

MAJOR COST COMPONENTS

The primary cost components in AI systems include training cost, inference cost, data storage, and monitoring and retraining, all of which significantly impact overall expenditure.

RECURRING INFERENCE COSTS

Large Language Models (LLMs) introduce additional ongoing costs due to high recurring inference demands, which can greatly influence operational budgets and resource allocation strategies.

Latency Consideration S

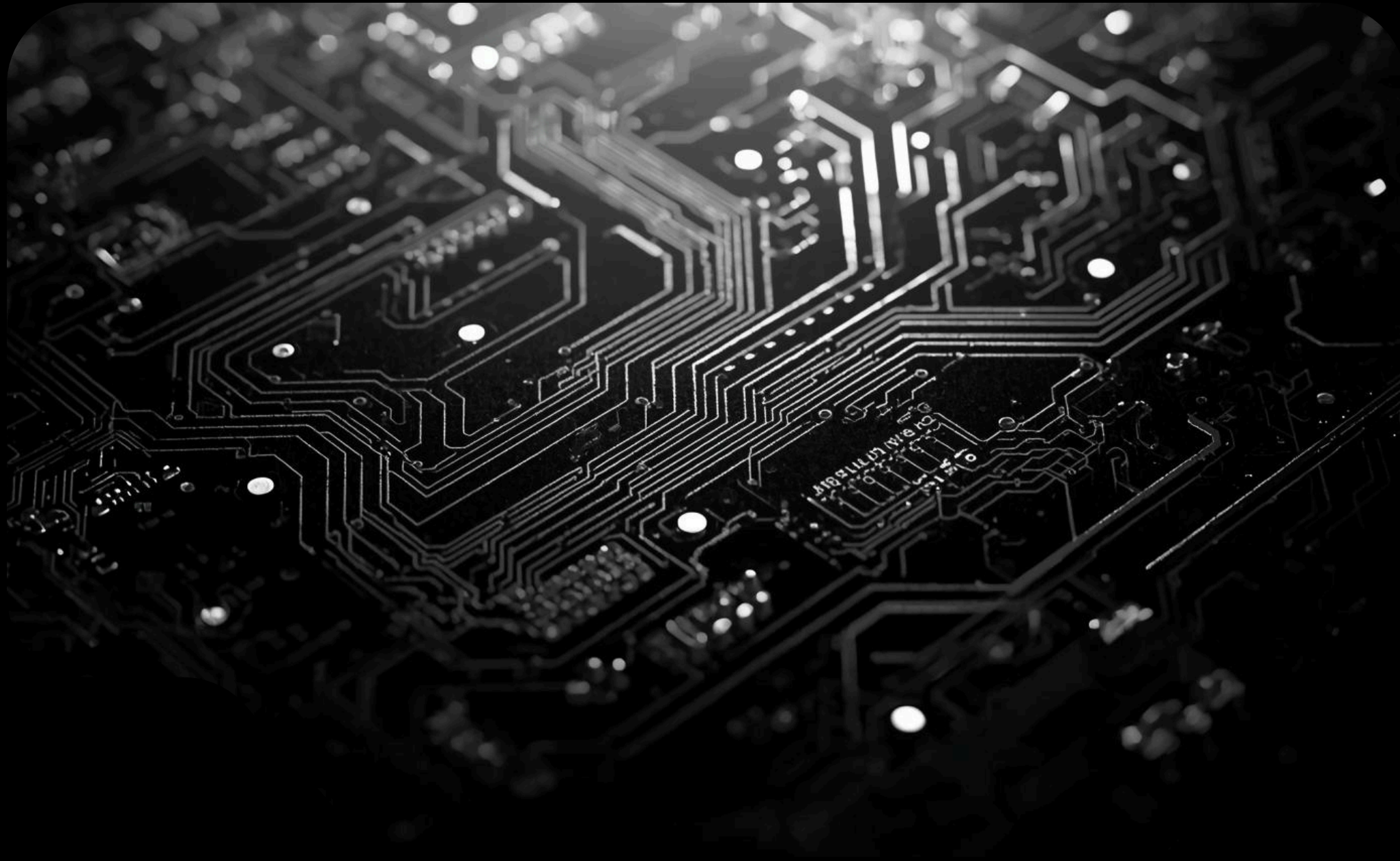
USER EXPERIENCE

Latency significantly affects user experience, as slower response times can lead to frustration and decreased satisfaction, ultimately impacting user retention and engagement with the system.

SYSTEM SCALABILITY

High latency can limit system scalability, as increased delays may hinder the ability to handle larger workloads, resulting in bottlenecks that affect overall performance and efficiency.

Architecture Trade-offs



Choices

- Scalability considerations
- Maintainability strategies
- Cost effectiveness

Examples

- Batch vs real-time inference
- Cloud API vs self-hosted model

Example Architecture

Exploring Chatbot Options and Trade-offs

LLM API

The LLM API is **fast to build** but incurs higher costs, making it suitable for rapid deployment.

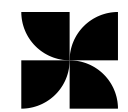
RAG-based Chatbot

A RAG-based chatbot offers a **balanced approach**, combining efficiency and cost-effectiveness for diverse applications in various environments.

Fine-tuned Local Model

While complex, a fine-tuned local model is **cheaper at scale**, providing an economical choice for high-volume usage.

Monitoring



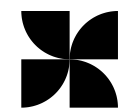
Low latency

Ensures quick response times, enhancing user experience and satisfaction with the system.



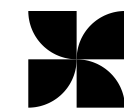
Predictable cost

Cost efficiency is achieved through minimized resource usage and well-defined pricing structures.



Easier monitoring

Simplifies performance assessments and allows for prompt identification of potential issues.



Classical ML

Utilizes established algorithms to streamline processes and improve accuracy in predictions.

Trade-off Matrix Analysis

Evaluating approaches on key metrics

No solution wins on all axes; it is essential to consider trade-offs in accuracy, cost, latency, and complexity when choosing an approach.



Trade-off Matrix

No solution wins on all axes; it is essential to consider trade-offs in accuracy, cost, latency, and complexity when choosing an approach.

Accuracy

No solution wins on all axes.

Cost

Assessing cost is critical to decision-making.

Latency

Timeliness impacts user experience and satisfaction.

Complexity

Simplifying architecture can enhance maintainability and performance.



Common Architecture Mistakes

LLMs

Choosing LLMs for everything can lead to inefficiencies and costs.

Cost

Ignoring cost at design time can result in budget overruns later.

Fallback

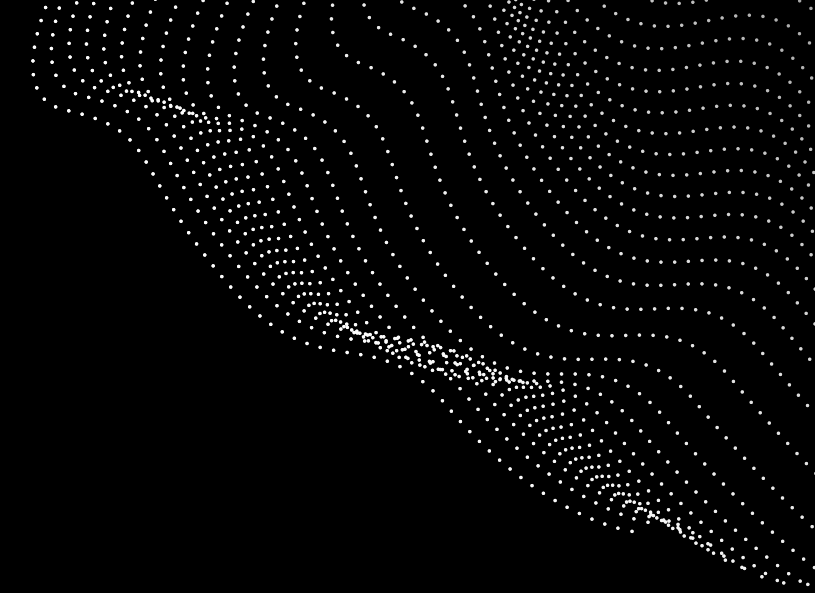
No fallback strategies can create risks during critical failures and errors.

Monitoring

No monitoring plan prevents early detection of potential issues.

Late Mistakes

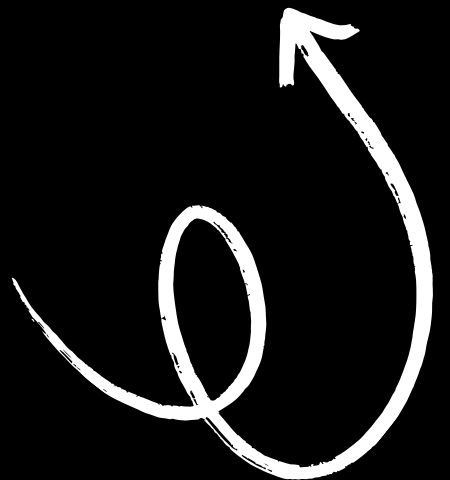
These mistakes surface late and are expensive to resolve.



Group Exercise

Decision Making in Practice

Given a problem, decide on an approach (Rules / ML / NLP / LLM), identify constraints, and sketch a high-level architecture. Focus on reasoning, not tools.



Key Takeaways

RIGHT APPROACH

Choosing the **right approach** matters more than the model itself. Effective decisions can dramatically influence the outcome of AI projects and their overall success.

ARCHITECTURE SUCCESS

Architecture decisions play a critical role in defining success. Well-planned frameworks and structures ensure that AI solutions are scalable, maintainable, and aligned with business goals.

LLMS POWER

Large Language Models (LLMs) are powerful but costly. Their implementation and operational expenses necessitate careful budgeting and resource allocation to maximize value and efficiency.

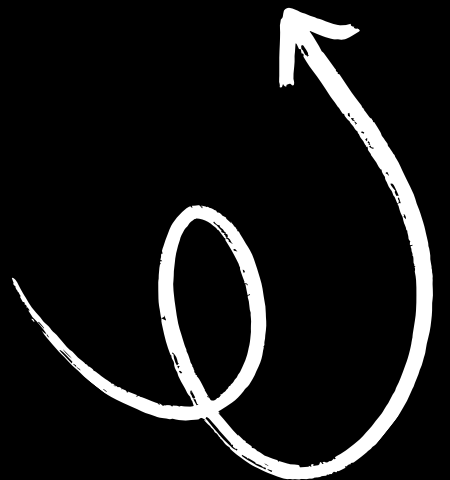
BALANCING TRADE-OFFS

Good AI design is about balancing trade-offs. Understanding the compromises between performance, cost, and complexity is essential for creating effective and sustainable AI systems.

Homework / Practice

Apply AI to a real problem

For one real problem, choose an AI approach, justify why alternatives are not suitable, estimate cost and latency, and sketch the proposed architecture for clarity.



Thank You

